Algorithm 1: $((\mathfrak{R}_g, \mathcal{D}_g, \mathcal{L}_g))_{g\in\mathcal{G}} \leftarrow \texttt{MultiGPUQueuedRollout}(\pi, \mathsf{m}, N)$

/* The rollout profile is frozen and identical on all GPUs; queues and rollout state are GPU-local. */

Parameters: finite PBG $(\mathcal{S}, \mathcal{T}, \mathcal{P}, o, (\mathcal{E}_\sigma)_\sigma, \mathcal{X}, \mu, \chi)$; frozen $\pi = \pi^{\theta_k}$; $\mathsf{m} \in \{\mathsf{normal}, \mathsf{RB}, \mathsf{branched\text{-}RB}\}$; GPU set $\mathcal{G}$; N sequential trajectory starts, unused in branched-RB

Parameters: finite execution-family poset $(\mathcal{M}, \le_{\mathcal{M}})$; public routing map $\kappa : \{\sigma : o(\sigma) \in \mathcal{P}\} \to \mathcal{M}$ satisfying $\sigma \preccurlyeq \tau \Rightarrow \kappa(\sigma) \le_{\mathcal{M}} \kappa(\tau)$ at player nodes; one curried forward implementation $\texttt{Forward}_m$ per m

Hyperparameters: $\beta : \mathcal{S} \setminus \mathcal{T}^{\mathsf{pre-c}} \to [1, \infty)$ for branched-RB; per-GPU hard memory limits; calibrated family packers and full-batch thresholds

Definitions: $\mathsf{seq} \leftarrow \mathbb{1}_{\mathsf{m}\neq\mathsf{branched\text{-}RB}}$; $\mathcal{T}_{\mathsf{m}} \leftarrow \mathcal{T}$ if $\mathsf{m} = \mathsf{normal}$, else $\mathcal{T}^{\mathsf{pre-c}}$; $\mathrm{Stop}_{\mathsf{m}}(\sigma) \leftarrow \mathbb{1}_{\sigma\in\mathcal{T}_{\mathsf{m}}}$

Definitions: $\mathrm{Round}(z; K) = \lfloor z \rfloor + \mathrm{Bernoulli}(z - \lfloor z \rfloor; K)$; $\iota(\mathsf{s})$ is the schedule-independent abstract address of s (trajectory index and path in sequential modes, abstract tree path in branched-RB); all draws use disjoint keys $K(k, g, \iota(\mathsf{s}), u, \mathsf{purpose})$

Definitions: $\mathbb{P}^{\mathfrak{r}_{\mathsf{s}}}(x \mid \sigma) = \dfrac{\mu(x)\prod_{q\in\mathcal{P}\cup\{\mathsf{c}\}} r_{\mathsf{s},q}(\sigma \mid x_q)}{\sum_{y\in\mathcal{X}} \mu(y)\prod_{q\in\mathcal{P}\cup\{\mathsf{c}\}} r_{\mathsf{s},q}(\sigma \mid y_q)}$

Definitions: $\texttt{Pack}_{g,m}(\mathcal{Q})$ returns a stable maximum-fill feasible occurrence batch; feasibility counts one model row per distinct exact model input, padded forward cost, child sampling, and expansion workspace; $\phi_{g,m}(\mathcal{B}) \in [0, 1]$ is its calibrated fill; $\mathrm{Full}_{g,m}(\mathcal{Q})$ holds when the packed batch reaches the family's full-batch threshold

Definitions: $\mathrm{Enqueue}_g(\mathcal{P}')$ appends each $\mathsf{s} \in \mathcal{P}'$ to the FIFO queue $\mathcal{Q}_{g,\kappa(\ell_g(\mathsf{s}))}$; $\mathrm{SafeAdmit}_g(a)$ reserves retained storage and the largest in-flight transaction before admitting a further sequential starts

Output: $\mathfrak{R}_g$: one labeled tree rooted at $\varnothing$ if branched-RB, or GPU-local sequential trajectory records otherwise; decision rows $\mathcal{D}_g$; stopped leaves $\mathcal{L}_g$

/* Poker specialization: $\mathrm{pre} \prec \mathrm{post}_{|\mathcal{P}|} \prec \cdots \prec \mathrm{post}_2$; street is packing metadata inside each post_A family. */

Deterministically partition $\{1, \ldots, N\} = \bigsqcup_{g\in\mathcal{G}} \mathcal{I}_g$ if $\mathsf{seq} = 1$

foreach $g \in \mathcal{G}$ ***in parallel*** **do**
- $\mathcal{D}_g, \mathcal{L}_g \leftarrow \varnothing$
- $(\mathcal{Q}_{g,m})_{m\in\mathcal{M}} \leftarrow \varnothing$
- **if** $\mathsf{seq} = 0$ **then**
  - $\mathfrak{S}_g \leftarrow \{\varnothing\}$
  - $\ell_g(\varnothing) \leftarrow \varnothing$
  - $\mathfrak{r}_{g,\varnothing} \leftarrow (1)_{q\in\mathcal{P}\cup\{\mathsf{c}\}}$
  - $\iota(\varnothing) \leftarrow \varnothing$
  - $\mathfrak{R}_g \leftarrow (\mathfrak{S}_g, \ell_g, N_g, \mathfrak{r}_g)$
  - $(\mathcal{P}_0, \mathcal{L}_0) \leftarrow \texttt{ChanceClose}_g(\{\varnothing\}, \mathsf{m})$
  - $\mathrm{Enqueue}_g(\mathcal{P}_0)$
  - $\mathcal{L}_g \leftarrow \mathcal{L}_g \sqcup \mathcal{L}_0$
  - $\mathcal{U}_g \leftarrow \varnothing$ /* exactly one root on this GPU */
- **else**
  - $\mathfrak{R}_g \leftarrow \varnothing$
  - $\mathcal{U}_g \leftarrow \mathcal{I}_g$ /* unstarted trajectory indices */
- **end**
- **while** $\mathcal{U}_g \neq \varnothing$ *or* $\bigsqcup_{m\in\mathcal{M}} \mathcal{Q}_{g,m} \neq \varnothing$ **do**
  - $\mathcal{F}_g \leftarrow \{m \in \mathcal{M} : \mathrm{Full}_{g,m}(\mathcal{Q}_{g,m})\}$
  - **if** $\mathcal{F}_g \neq \varnothing$ **then**
    - $m \leftarrow \arg\max_{m\in\min_{\le_{\mathcal{M}}}\mathcal{F}_g} (\mathrm{load}(\mathcal{Q}_{g,m}), \mathrm{age}(\mathcal{Q}_{g,m}))$ /* lexicographic; upstream full work first */
  - **else if** $\mathsf{seq} = 1$ *and* $a_g \leftarrow \max\{a \le |\mathcal{U}_g| : \mathrm{SafeAdmit}_g(a)\} > 0$ **then**
    - Remove a_g indices i from $\mathcal{U}_g$ in stable order and, in one device batch, allocate trajectory-start occurrences s_i^0 with $\ell_g(\mathsf{s}_i^0) = \varnothing$, $\mathfrak{r}_{g,\mathsf{s}_i^0} = (1)_q$, $X_i \sim \mu$ using $K(k, g, i, 0, \mathsf{private})$, and $\iota(\mathsf{s}_i^0) = (i, \varnothing)$
    - Append these occurrences to $\mathfrak{R}_g$
    - $(\mathcal{P}_0, \mathcal{L}_0) \leftarrow \texttt{ChanceClose}_g(\{\mathsf{s}_i^0\}_i, \mathsf{m})$
    - $\mathrm{Enqueue}_g(\mathcal{P}_0)$
    - $\mathcal{L}_g \leftarrow \mathcal{L}_g \sqcup \mathcal{L}_0$
    - **continue**
  - **else**
    - $\mathrm{assert}(\bigsqcup_m \mathcal{Q}_{g,m} \neq \varnothing)$
    - $m \leftarrow \arg\max_{m:\mathcal{Q}_{g,m}\neq\varnothing} (\phi_{g,m}(\texttt{Pack}_{g,m}(\mathcal{Q}_{g,m})), \mathrm{age}(\mathcal{Q}_{g,m}), -h(m))$ /* forced partial drain; h is any topological height */
  - **end**
  - $\mathcal{B} \leftarrow \texttt{Pack}_{g,m}(\mathcal{Q}_{g,m})$ and remove its occurrences from $\mathcal{Q}_{g,m}$
  - Let $\mathcal{V} \subseteq \mathcal{B}$ contain one stable representative of each distinct exact model input $(\ell_g(\mathsf{s}), \mathfrak{r}_{g,\mathsf{s}})$
  - **foreach** $\mathsf{s} \in \mathcal{V}$ ***in one family-m GPU call*** **do**
    - $\sigma_{\mathsf{s}} \leftarrow \ell_g(\mathsf{s})$
    - $p_{\mathsf{s}} \leftarrow o(\sigma_{\mathsf{s}})$
    - $\Pi_{\mathsf{s}} \leftarrow \texttt{Forward}_m(\sigma_{\mathsf{s}}, \mathfrak{r}_{g,\mathsf{s}}; \theta_k) = \big(\pi(\cdot \mid x_{p_{\mathsf{s}}}, \sigma_{\mathsf{s}}), V^{\theta_k}(x_{p_{\mathsf{s}}}, \sigma_{\mathsf{s}}), \ldots\big)_{x_{p_{\mathsf{s}}}\in\mathcal{X}_{p_{\mathsf{s}}}}$
  - **end**
  - Gather Π_{s} from representatives to all $\mathsf{s} \in \mathcal{B}$
  - $(\mathcal{A}, \mathcal{F}) \leftarrow \texttt{SampleChildren}_g(\mathcal{B}, (\Pi_{\mathsf{s}})_{\mathsf{s}\in\mathcal{B}}, \mathsf{m})$
  - $(\mathcal{P}', \mathcal{L}') \leftarrow \texttt{ChanceClose}_g(\mathcal{F}, \mathsf{m})$
  - $\mathcal{D}_g \leftarrow \mathcal{D}_g \sqcup \mathcal{A}$
  - $\mathrm{Enqueue}_g(\mathcal{P}')$
  - $\mathcal{L}_g \leftarrow \mathcal{L}_g \sqcup \mathcal{L}'$
  - $\mathrm{assert}(\bigsqcup_m \mathcal{Q}_{g,m} = \{\text{all live, non-in-flight player occurrences on GPU } g\})$
  - assert(no chance occurrence is pending)
  - assert(hard structural and memory budgets hold)
- **end**
- **if** $\mathsf{seq} = 0$ **then**
  - assert($\mathfrak{S}_g$ is one tree with the unique root $\varnothing$)
- **end**

end

Barrier($\mathcal{G}$) once after all local rollout outputs are complete; perform no queue or model collective inside the rollout loop

return $((\mathfrak{R}_g, \mathcal{D}_g, \mathcal{L}_g))_{g\in\mathcal{G}}$

Algorithm 2: $(\mathcal{A}, \mathcal{F}') \leftarrow \texttt{SampleChildren}_g(\mathcal{B}, \Pi, \mathsf{m})$

/* $\mathcal{B}$ is a player batch with curried output Π, or a chance batch with $\Pi = \bot$. */

Input: sampled occurrences $\mathcal{B} = (\mathsf{s}_j)_{j=1}^b$ of one owner kind; mode m; GPU-local rollout state and schedule-independent keys

Output: decision rows $\mathcal{A}$ (empty for chance); all sampled children $\mathcal{F}'$

$\mathcal{A}, \mathcal{F}' \leftarrow \varnothing$

for $j = 1, \ldots, b$ ***in one device batch*** **do**
- $\sigma_j \leftarrow \ell_g(\mathsf{s}_j)$
- $q_j \leftarrow o(\sigma_j)$
- $n_j \leftarrow \begin{cases} 1, & \mathsf{m} \neq \mathsf{branched\text{-}RB}, \\ \mathrm{Round}(\beta(\sigma_j); K(k, g, \iota(\mathsf{s}_j), 0, \mathsf{round})), & \mathsf{m} = \mathsf{branched\text{-}RB}; \end{cases}$
- **if** $\mathsf{m} = \mathsf{branched\text{-}RB}$ **then**
  - $N_g(\mathsf{s}_j) \leftarrow n_j$
- **end**

end

$\mathcal{I} \leftarrow \{(j, u) : 1 \le j \le b, 1 \le u \le n_j\}$ by prefix sum

reserve the exact child/edge block and maximum-feasible ragged sampler workspace

fail loudly rather than changing $(n_j)_j$ if a hard budget is exceeded

foreach $(j, u) \in \mathcal{I}$ ***in maximum-feasible device batches*** **do**
- $X_{j,u} \leftarrow \begin{cases} X_{i(j)}, & \mathsf{m} \neq \mathsf{branched\text{-}RB}, \\ X \sim \mathbb{P}^{\mathfrak{r}_{g,\mathsf{s}_j}}(\cdot \mid \sigma_j) \text{ with } K(k, g, \iota(\mathsf{s}_j), u, \mathsf{posterior}), & \mathsf{m} = \mathsf{branched\text{-}RB}; \end{cases}$
- $E_{j,u} \sim \begin{cases} \pi(\cdot \mid X_{j,u,q_j}, \sigma_j) \text{ from } \Pi_j, & q_j \in \mathcal{P}, \\ \chi(\cdot \mid X_{j,u}, \sigma_j), & q_j = \mathsf{c}, \end{cases}$ using the corresponding action or chance key
- allocate a distinct abstract child $\mathsf{s}_j\mathsf{a}_{j,u}$ and set $\iota(\mathsf{s}_j\mathsf{a}_{j,u}) = (\iota(\mathsf{s}_j), u)$, $\ell_g(\mathsf{s}_j\mathsf{a}_{j,u}) = \sigma_j E_{j,u}$
- $\mathfrak{r}_{g,\mathsf{s}_j\mathsf{a}_{j,u},v} gets \mathfrak{r}_{g,\mathsf{s}_j,v}$ for $v \neq q_j$
- $\mathfrak{r}_{g,\mathsf{s}_j\mathsf{a}_{j,u},q_j} \leftarrow \mathrm{Normalize}\Big(\mathfrak{r}_{g,\mathsf{s}_j,q_j} \odot \begin{cases} \pi(E_{j,u} \mid -, \sigma_j), & q_j \in \mathcal{P}, \\ \chi(E_{j,u} \mid -, \sigma_j), & q_j = \mathsf{c} \end{cases}\Big)$
- append $\mathsf{s}_j\mathsf{a}_{j,u}$ to the local rollout record and $\mathcal{F}'$

end

if $\Pi \neq \bot$ **then**
- $\mathcal{A} \leftarrow \{(\mathsf{s}_j, (\mathsf{a}_{j,u}, E_{j,u})_{u=1}^{n_j}, n_j, \mathfrak{r}_{g,\mathsf{s}_j}, \mathrm{Old}(\Pi_j, (E_{j,u})_{u=1}^{n_j})) : 1 \le j \le b\}$ /* retain only loss-required old policy/value slices */

end

Discard branched-RB posterior profiles $X_{j,u}$

return $(\mathcal{A}, \mathcal{F}')$

Algorithm 3: $(\mathcal{P}', \mathcal{L}') \leftarrow \texttt{ChanceClose}_g(\mathcal{F}, \mathsf{m})$

/* Close every chance-only segment before returning control to the model-family scheduler. */

Input: new sampled occurrences $\mathcal{F}$; mode m; GPU-local rollout state

Output: next player occurrences $\mathcal{P}'$ and stopped leaves $\mathcal{L}'$

$\mathcal{P}', \mathcal{L}' \leftarrow \varnothing$

while $\mathcal{F} \neq \varnothing$ **do**
- Partition $\mathcal{F}$ into $\mathcal{L}_0 = \{\mathsf{s} : \mathrm{Stop}_{\mathsf{m}}(\ell_g(\mathsf{s})) = 1\}$, $\mathcal{P}_0 = \{\mathsf{s} : \mathrm{Stop}_{\mathsf{m}}(\ell_g(\mathsf{s})) = 0, o(\ell_g(\mathsf{s})) \in \mathcal{P}\}$, and $\mathcal{C}_0 = \{\mathsf{s} : \mathrm{Stop}_{\mathsf{m}}(\ell_g(\mathsf{s})) = 0, o(\ell_g(\mathsf{s})) = \mathsf{c}\}$
- $\mathrm{assert}(|\mathcal{F}| = |\mathcal{L}_0| + |\mathcal{P}_0| + |\mathcal{C}_0|)$
- $\mathcal{P}' \leftarrow \mathcal{P}' \sqcup \mathcal{P}_0$
- $\mathcal{L}' \leftarrow \mathcal{L}' \sqcup \mathcal{L}_0$
- **if** $\mathsf{m} = \mathsf{branched\text{-}RB}$ **then**
  - $N_g(\mathsf{s}) \leftarrow 0$ for $\mathsf{s} \in \mathcal{L}_0$
- **end**
- Partition $\mathcal{C}_0$ into stable maximum-feasible chance-sampling batches $(\mathcal{C}_v)_v$
- $\mathcal{F} \leftarrow \varnothing$
- **foreach** $\mathcal{C}_v$ **do**
  - $(\varnothing, \mathcal{F}_v) \leftarrow \texttt{SampleChildren}_g(\mathcal{C}_v, \bot, \mathsf{m})$
  - $\mathcal{F} \leftarrow \mathcal{F} \sqcup \mathcal{F}_v$
- **end**

end

return $(\mathcal{P}', \mathcal{L}')$